K.Ramakrishnan College of Technology
Autonomous
Affiliated to Anna University Chennai, Approved by AICTE New Delhi,
Accredited by NBA and with 'A+' grade by NAAC
Samayapuram, Tiruchirappalli - 621 112, Tamilnadu, India.
Incubating Minds, Catalyzing Careers

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

20CS7503 & DESIGN PROJECT – III

END SEMESTER PRACTICAL EXAMINATION- NOV / DEC 2025

REVIEW PRESENTATION

Batch No.: 17

Date: 1.12.2025

Session: AN

BIOMETRIC ATTENDANCE USING SMARTPHONE AND GPS

Guide Name:

Mr.M.A.Prasanna B.E,M.Tech

Assistant Profosser

Team Members:

Akalya.R (811722104007)

Elakkiya.S (811722104038)

Janaani.S.V (811722104059)

OBJECTIVE OF THE PROJECT

1.The main objective is to develop a smart and secure attendance system that uses fingerprint authentication and GPS location verification on smartphones.

2.Employees can mark attendance using their smartphones without the need for extra hardware.

3.Use GPS to ensure attendance is marked only from authorized locations.

INTRODUCTION

1.Biometric Attendance Using Smartphone and GPS Coordinates is an innovative solution aimed at modernizing and securing the traditional attendance system by leveraging biometric verification and real-time location tracking.

2.In educational institutions and workplaces, accurate attendance monitoring is essential for ensuring discipline, productivity, and accountability.

3.However, conventional systems such as manual registers or card-based methods are prone to manipulation, proxy attendance, and inefficiency.

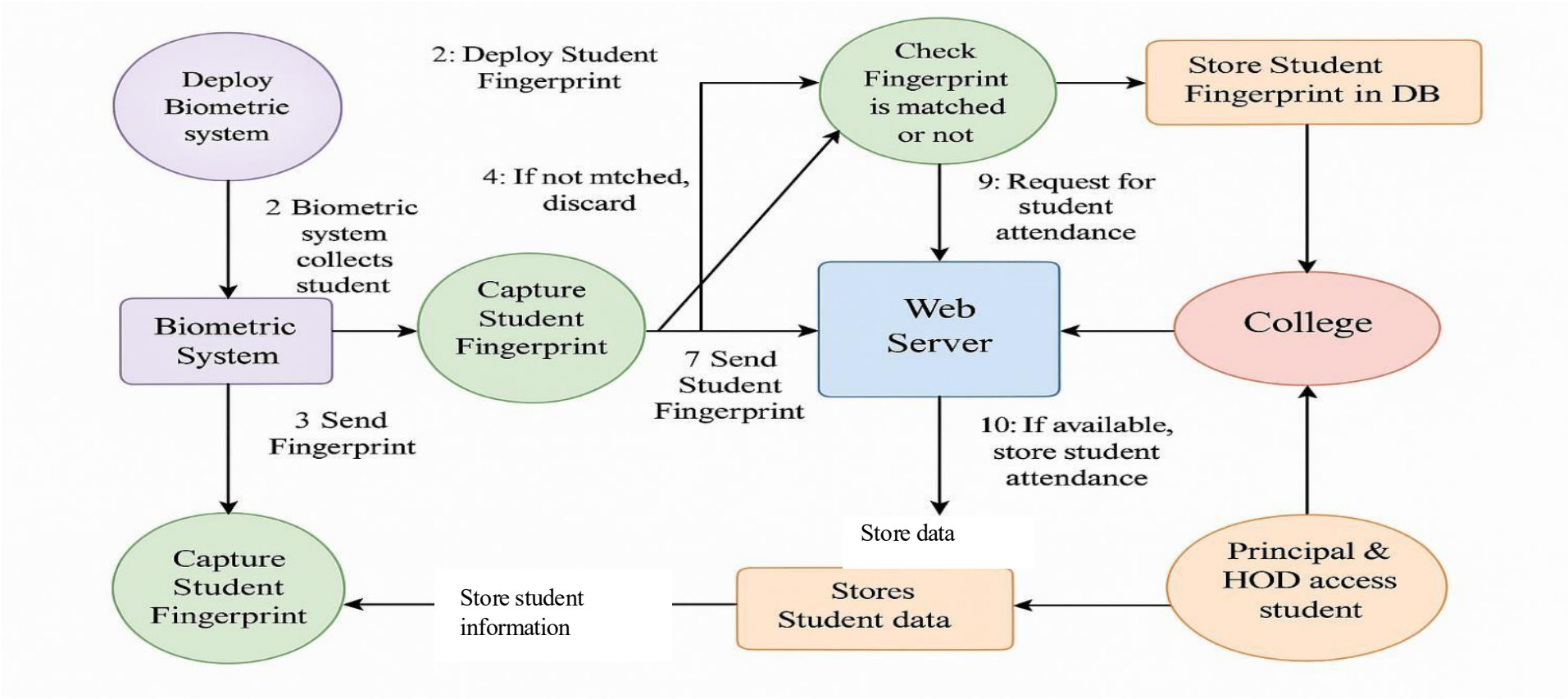
4.The system functions by first verifying the identity of the user through fingerprint scanning using the smartphone's built-in biometric sensor.

5.Once the user is authenticated, the application captures the real-time GPS coordinates of the device.

LITERATURE SURVEY

TITLE OF PAPER	AUTHOR(S)	PAPER GIST	TECHNOLOGY USED
Smartphone-Based Multi-Modal Biometric Authentication	A.Kumar , R.Singh	Combines fingerprint and face recognition on smartphones with GPS.	Android SDK . Fingerprint API,Face Recognition GPS Services
Location-Aware Biometric Security System	S.Banerjee , M.Rao	Uses biometric login on user's GPS location matches Zones.	GPS Geofencing,Biometric API,Secure Enclave
Behavioural Biometrics with GPS for Mobile Authentication	L. Wang , T.Chen	Uses touch dynamics and movement patterns to verify user.	Touch Biometrics , Machine Learning , GPS Tracking
GPS-Enchanced Two-Factor Biometric Authentication	K.Prakash , V.Nair	Integrates biometric Scan and GPS location as second factor	Fingerprint Scanner , GPS Module, 2FA Framework
Continuous Biometric Authentication via Smartphone Sensors	J.Patel , H.Verma	Uses accelometer and GPS Screen interaction.	Sensor Fusion , GPS ML Algorithms
Mobile Identity Verification Using Smartphone	D.Shah , N.Babu	Ensures secure transactions by cross checking	GPS validation,Finferprint Recognition

EXISTING SYSTEM ARCHITECTURE



PROBLEM IDENTIFICAITON

1. Multiple user cannot log in single device.

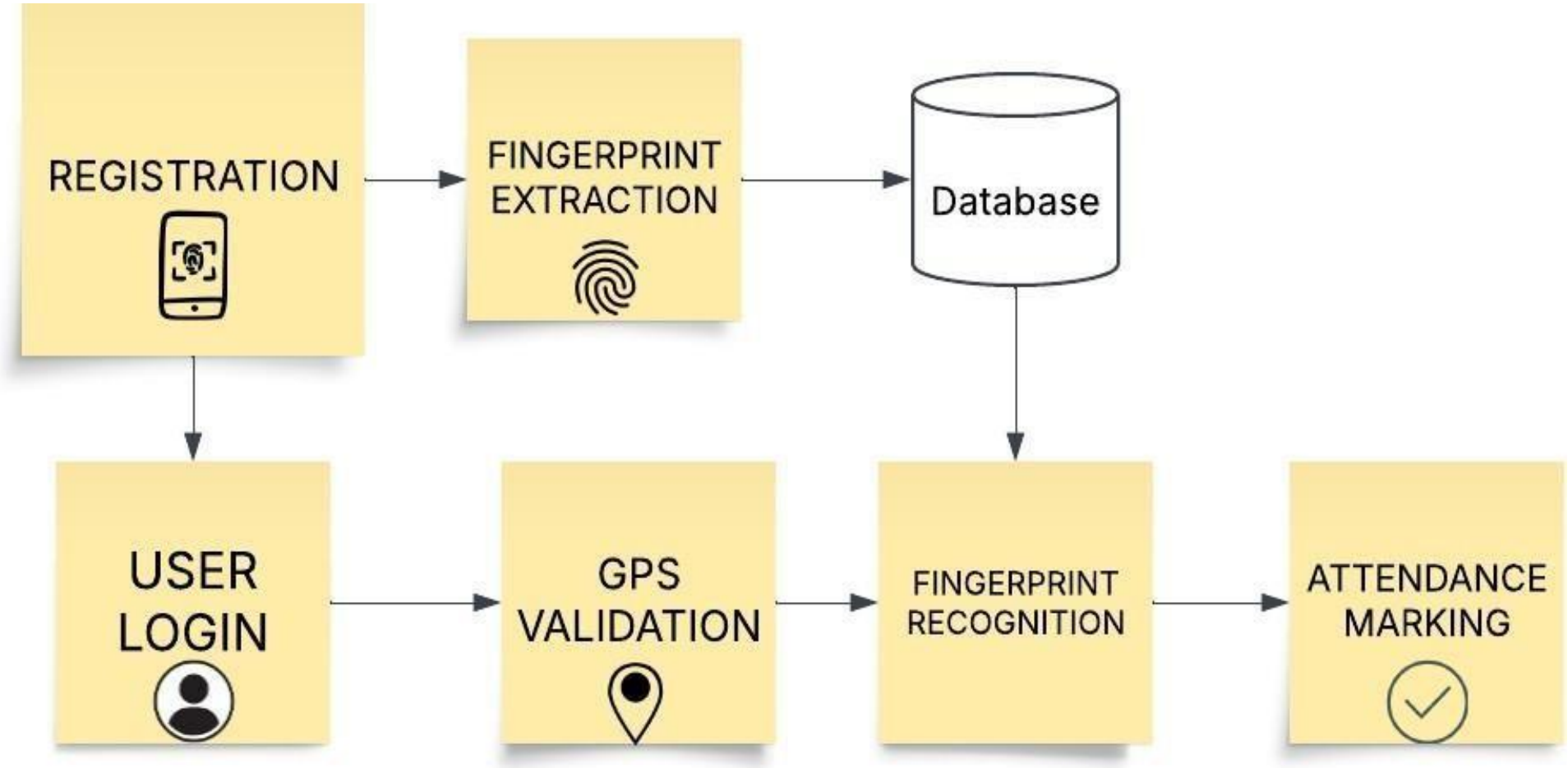
2. Gradle issues

3. Not applicable for iphone

4. Issues with recognition of damaged fingerprints

5. Mobile device compatibility

PROPOSED SYSTEM ARCHITECTURE



SOFTWARE AND HARDWARE REQUIREMENTS

HARDWARE

1. Window 10 OS
2. 8GB RAM
3. Intel i5 processor

SOFTWARE

1. Android studio
2. Java SDK
3. Graddle

MODULES

1. **User Registration**

2. **User Login**

3. **GPS Validation**

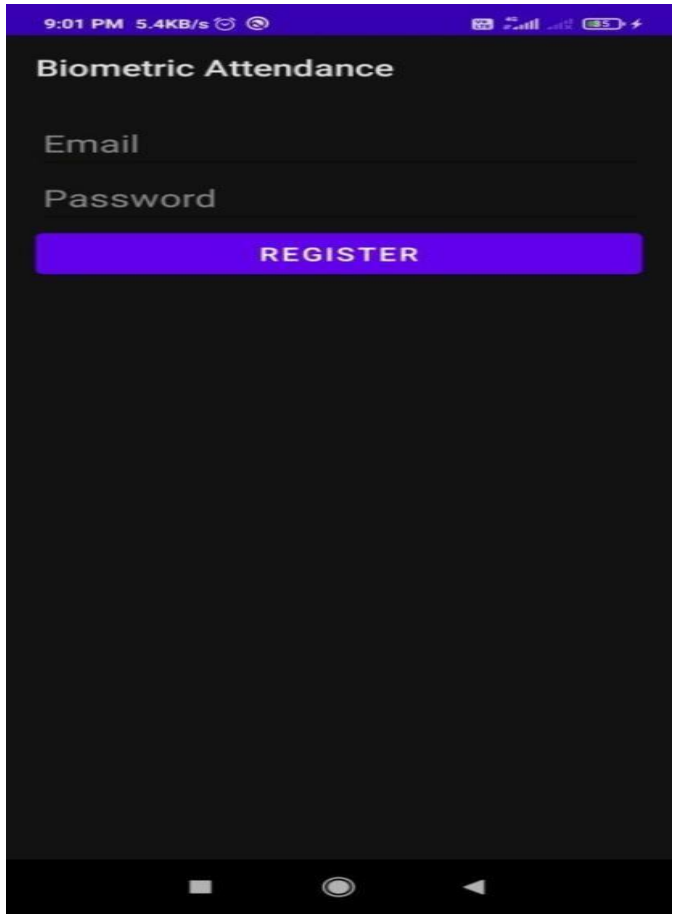
4. **Fingerprint Authentication**

5. **Attendance marking**

IMPLEMENTATION OF MODULE-1

1.New users must first register in the system by providing personal details, and scanning their fingerprint.

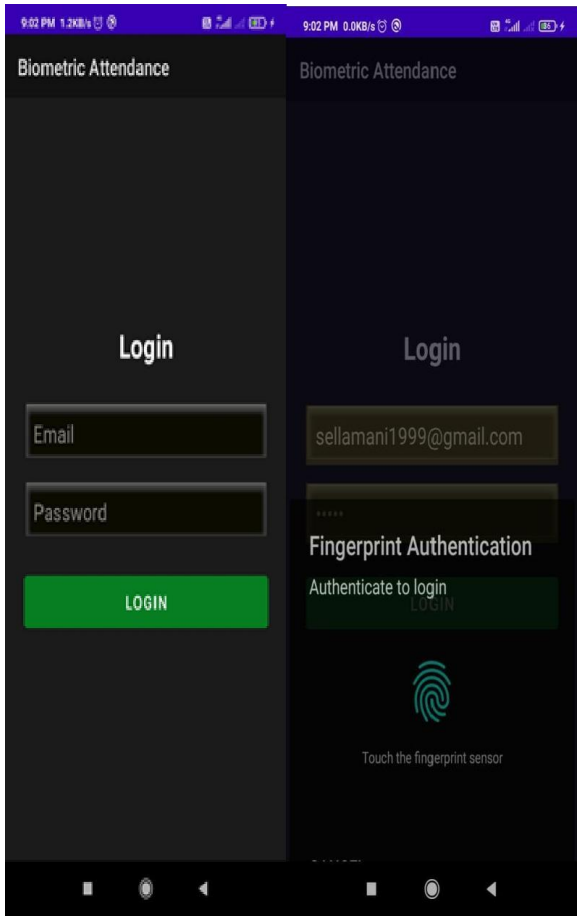
2.The user's fingerprint is captured using a fingerprint scanner. The system will save these biometric templates securely in the database for later verification.



IMPLEMENTATION OF MODULE-2

1.Users (employees, students, etc.) can log in to the system using their credentials (username, password) or through biometric authentication like fingerprint recognition.

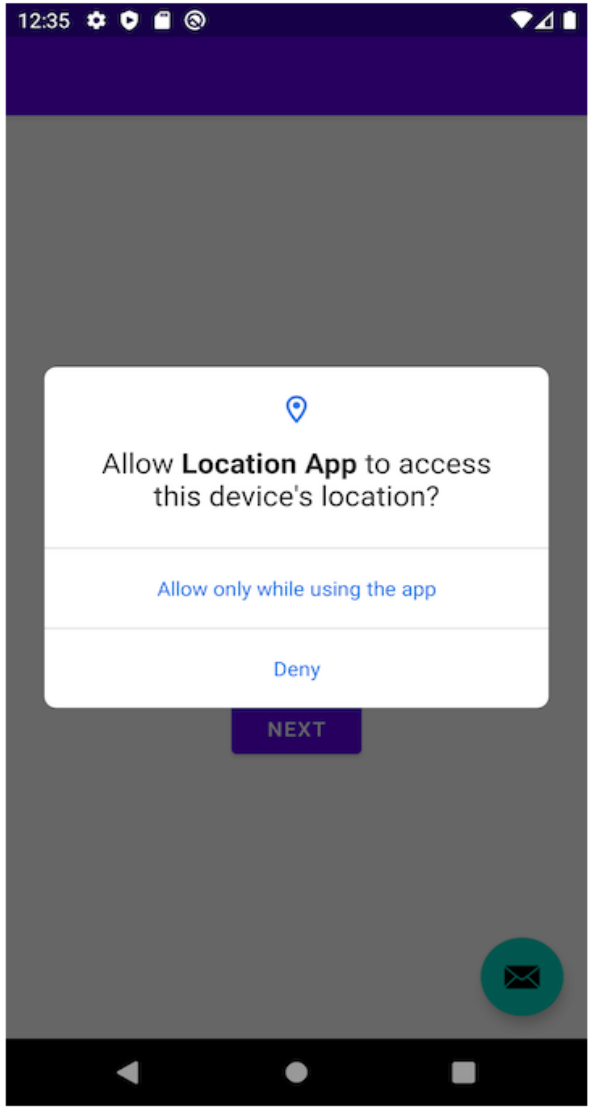
2.A login page where users enter their username and password. For biometric login, the system will use fingerprint to authenticate the user, Ensure passwords are encrypted, and biometric data is securely stored.



IMPLEMENTATION OF MODULE-3

1.The system checks whether the employee or user is at the assigned location (office, worksite, etc.) using GPS coordinates.

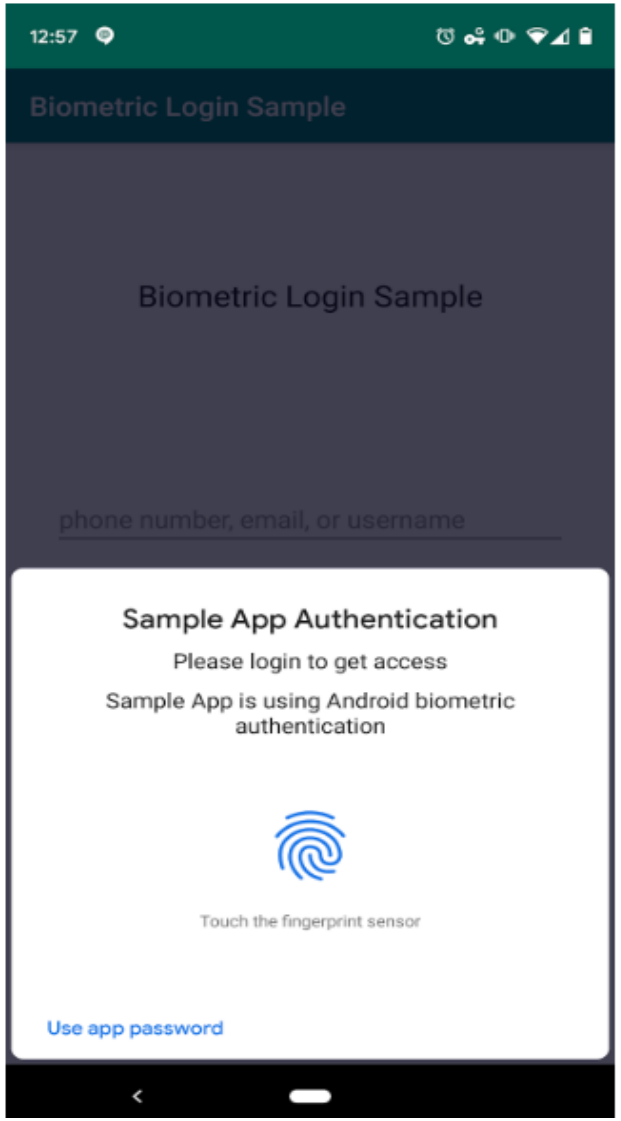
2.The system captures the users GPS coordinates using the mobile device or a GPS enabled device.



IMPLEMENTATION OF MODULE-4

1.If the Fingerprint matchs the registered fingerprint, attendance is marked as valid.

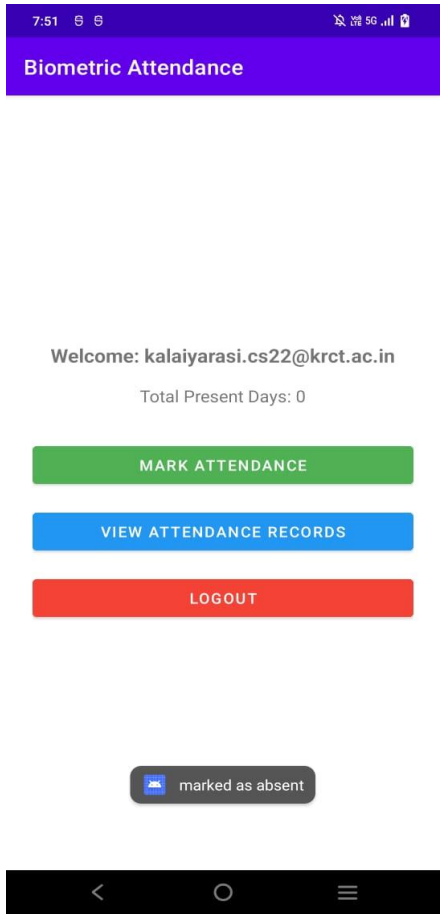
2.Attendance data is sent to a cloud or local database for record-keeping.

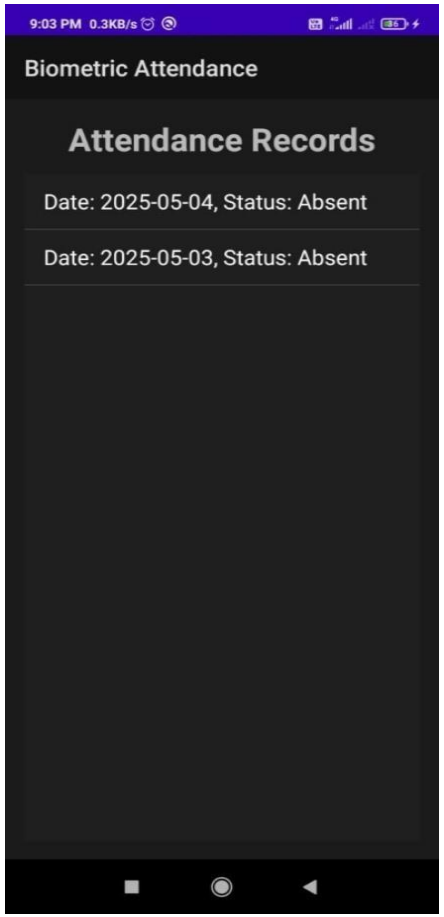


IMPLEMENTATION OF MODULE-5

1. If the GPS coordinates match the allowed area, attendance is marked as valid.

2. Attendance data is sent to a cloud or local database for record-keeping.





CONCLUSION

1.In conclusion, the Biometric Attendance Using Smartphone and GPS Coordinates project provides a modern, secure, and efficient solution for managing attendance by combining the strengths of biometric authentication and real-time GPS location tracking.

2. It also makes it simple and convenient to use.

FUTURE ENHANCEMENTS

1.One key improvement is the implementation of multiple user logins on the same device, which would be particularly beneficial in shared environments such as classrooms, offices, or laboratories where multiple individuals use a single smartphone or tablet.

2.Make it available for iphone users by adding face recognition feature.